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CT-3

(1) $ax(1-x)y'' - 12y' + 4y = 0$

$$P(x) = \frac{-12}{ax(1-x)} = \frac{-4}{3x(1-x)} = \cancel{\frac{4}{3x(x-1)}} \quad \frac{4}{3x(x-1)}$$

$$Q(x) = \frac{4}{ax(1-x)}$$

~~P(x)~~ $\cancel{\longrightarrow \text{Pole}}$ $x = 0, 1 \longrightarrow$ Regular Singular Points.

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y'(x) = \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}$$

$$y''(x) = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Substituting into the ODE :

$$a \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-1} - a \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r}$$
$$- 12 \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} + 4 \sum_{n=0}^{\infty} a_n x^{n+r} = 0.$$

Lowest power of x : x^{r-1} @ $n=0$. Indicial Eqn :

~~$r(r-1) = 0$~~ $\cancel{\Rightarrow r = 0, 1}$

$$ar(r-1) - 12(r-1) = 0.$$

$$\Rightarrow ar^2 - 21r = 0 \Rightarrow \boxed{r = 0, \frac{7}{3}}$$

$$\cancel{9\sum_{n=1}^{\infty}} \quad \cancel{\sum_{n=0}} \quad 9\sum_{n=0}^{\infty}(n+r)(n+r-1)\,a_{n+1}\,x^{n+r} \rightarrow (i)$$

$$-9\sum_{n=0}^{\infty}(n+r)(n+r-1)\,a_n\,x^{n+r} - 12\sum_{n=0}^{\infty}(n+r+1)\,a_{n+1}\,x^{n+r} + 4\sum_{n=0}^{\infty}a_n x^{n+r} = 0 \quad \text{— (ii)}$$

[$n \rightarrow n+1$ for (i) & (ii)]

We do not ~~change~~ shift the index n as (i) & (ii) are 0 for $n=0$ $\longrightarrow$ Indicial eqn.

Comparing both sides:

$$9(n+r)\left[(n+r+1)\,a_{n+1} - (n+r-1)\,a_n\right] - 12(n+r+1)\,a_{n+1} + 4a_n = 0.$$

$$\Rightarrow \quad a_{n+1}\left[(n+r+1)\left\{9(n+r) - 12\right\}\right] + a_n\left[4 - 9(n+r)(n+r-1)\right] = 0$$

$$\Rightarrow \quad \boxed{a_{n+1} = \frac{\left[9(n+r)(n+r-1) - 4\right]}{9(n+r+1)\left[9(n+r) - \frac{12}{3}\right]}\,a_n}$$

For $r=0$:

$$a_{n+1} = \frac{\left[9n(n-1) - 4\right]}{9(n+1)(n-4/3)}\,a_n$$

$n=0$: $\quad a_1 = \dfrac{-4}{9\left(-\frac{4}{3}\right)}\,a_0 = \dfrac{a_0}{3}.$

$n=1$: $\quad a_2 = \dfrac{+4^{2}}{3\cancel{9}\cdot 2\cdot\left(+\frac{1}{\cancel{3}}\right)}\,a_1 = \dfrac{2a_1}{3} = \dfrac{2a_0}{9}$

$n=2$: $\quad a_3 = \dfrac{9\cdot 2 - 4}{9\cdot\cancel{3}\cdot\frac{2}{\cancel{3}}}\,a_2 = \dfrac{\cancel{14}\,7}{\cancel{18}\,9}\,a_2 = \dfrac{7}{9}\cdot\dfrac{2a_0}{9} = \dfrac{14}{81}\,a_0$

$$\therefore \quad y(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$\boxed{y_1(x) = a_0\left[1 + \frac{x}{3} + \frac{2x^2}{9} + \frac{14}{81}x^3 + \dots\right]} \quad \text{(Ans)}$$

$x=1$ is a singular point. The expansion is around $x=0$.

$\therefore$ $\boxed{R=1}$ — radius of convergence. The series is valid for $|x|<1$.

(4) $\quad xJ_n' = nJ_{n-1} - xJ_{n+1} \Rightarrow -xJ_0' = +xJ_1 \quad [n=0]$. — (i)

$xJ_n' = xJ_{n-1} - nJ_n \Rightarrow xJ_0' = xJ_{-1} \quad [n=0]$. — (ii)

(i) + (ii) $\Rightarrow x\left[J_1 + J_{-1}\right] = 0$.

$\therefore$ This is true $\forall x$; $\boxed{J_1 + J_{-1} = 0}$ (Proved)

③ Generating function for Bessel's function:

$$\exp\left[\frac{x}{2}\left(t-\frac{1}{t}\right)\right] = \sum_{n=-\infty}^{\infty} t^n J_n(x)$$

Putting $t = i$:

LHS: $\exp\left[\frac{x}{2}\left(i-\frac{1}{i}\right)\right]$

$= \exp\left[\frac{x}{2} \cdot 2i\right]$

$= e^{ix} = \cos x + i\sin x.$

RHS: $\sum_{n=-\infty}^{\infty} i^n J_n(x)$

$\cancel{\ldots + J_{-4} + i^{-3} J_{-3} - J_{-2} + iJ_{-1} + J_0}$

$\cancel{+ iJ}$

$= \ldots + J_{-4} + i^{-3} J_{-3} - J_{-2} + i^{-1} J_{-1} + J_0$

$+ iJ_1 - J_2 + i^3 J_3 + J_4 + \ldots$

$= \ldots + J_{-4} + iJ_{-3} - J_{-2}$

$- iJ_{-1} + J_0 + iJ_1 - J_2 + - iJ_3 + J_4$

$+ \ldots$

$= \ldots + [(J_{-4} + J_4) - (J_{-2} + J_2)]$

$\cancel{+ \ldots]}$ $+ J_0$

$+ i[(J_{-3} - J_3) - (J_{-1} - J_1)] +$

$\ldots$

Now, $J_{-n} = (-1)^n J_n$

$\therefore$ RHS: $J_{-4} = J_4$; $J_{-3} = -J_3$, & so on.

RHS: $\ldots + [2J_4 - 2J_2 + J_0] +$

$i[2J_3 - 2J_1] + \ldots$

Comparing the real part of both LHS & RHS:

$$\boxed{\cos x = J_0 - 2J_2 + 2J_4 - \ldots}$$ (Proved)